Decentralized Signal Detection via Expectation Propagation Algorithm for Uplink Massive MIMO Systems

Zhenyu Zhang, Hua Li, Yuanyuan Dong, Xiyuan Wang, and Xiaoming Dai

***Abstract*—The existing centralized antenna array architecture and the corresponding centralized baseband processing based signal detection algorithms incur huge burden on the raw baseband data rates and chip input/output bandwidth for massive multiple-input multiple-output (MIMO) systems. In this paper, we propose a novel decentralized baseband processing based signal detection under the framework of the expectation propagation algorithm (EPA) for uplink massive MIMO systems. This architecture partitions the base station antennas into multiple independent antenna clusters, each associated with analog and digital modulation circuitry, independent radio-frequency chains, and computing hardwares. The antenna clusters only access local channel state information and require low communication bandwidth. In the proposed scheme, the decentralized processing unit executes local channel estimation and signal detection, and delivers the partial message to the centralized processing unit parallelly, where the message contains the local means and variances iteratively updated by the EPA detector from each antenna cluster. The centralized processing unit operates the message fusion based on the proposed message fusion rule for decoding operation. Simulation results illustrate that the proposed decentralized EPA signal detection approaches the conventional centralized one with marginal computational complexity overhead when the number of antenna clusters is small, and outperforms the state-of-the-art decentralized counterparts.**

***Index Terms*—Massive multiple-input multiple-output (MIMO), antenna cluster, decentralized signal detection, expectation propagation algorithm (EPA), message fusion.**

I. Introduction

MASSIVE multiple-input multiple-output (MIMO) system is a critical technology for the fifth generation (5G) wireless communication systems [1], [2] due to its significant gains in spectral efficiency, energy efficiency, link reliability, latency, and coverage compared with the traditional MIMO system [3]. Massive MIMO systems deploy tens to hundreds of active antenna elements at the base station (BS) to simultaneously serve tens of user equipments in the same frequency resource. However, the deployments of practical massive MIMO systems with large-scale antenna arrays and the BS centralized baseband processing result in huge implementation challenges, such as the excessively high computational complexity, raw baseband data rates, and chip input/output bandwidth. The excessively high amount of raw baseband date must be transferred between the baseband processing unit and the radio-frequency (RF) chains at the BS [4]. For example, in a 256-antennas massive MIMO system with 80 MHz bandwidth and 12-bit analog-to-digital converters, the raw baseband data rates exceed 1 Tbit/s. Such high data rates result in raw baseband data stream exceeding the bandwidth of the existing high-speed interconnects, such as the common public radio interface (CPRI), and place huge burden on the hardware implementations in practice [4].

In practical massive MIMO systems, the existing signal detection schemes are normally relied on centralized baseband processing in a single computing fabric, such as linear zero-forcing (ZF) and minimum mean square error (MMSE) detectors. These centralized detection schemes require the full channel state information (CSI) and full received signal at the BS, which results in high-rate raw baseband data streams sampled from large amount of RF chains at the BS and the centralized processing unit must transmit or process these data streams. To solve the above-mentioned issues, decentralized baseband processing architecture for the massive MIMO systems has been proposed recently [4]–[10]. In the decentralized massive MIMO systems, the large-scale antennas embedded on a single BS are partitioned into multiple independent antenna clusters. Each antenna cluster is associated with a decentralized processing unit which includes the local channel estimation (CHEST) module and local detector. A centralized processing unit then fuses the message generated by each antenna cluster and delivers the fused message to the decoder for decoding. The decentralized scheme provides a flexible and scalable antenna architecture in which the antennas of each cluster can be flexibly allocated, rather than the complete system redesign for the conventional centralized massive MIMO architecture when increasing the number of antennas at the BS [11].

Various decentralized baseband processing based signal detection algorithms [4]–[10] have been proposed recently to mitigate the bandwidth bottlenecks of centralized massive MIMO architectures, and enable parallel baseband processing across BS antennas on multiple computing fabrics [10]. In

Manuscript received November 4, 2019; revised May 22, 2020; accepted July 1, 2020. Date of publication July 13, 2020; date of current version October 22, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61871029, and in part by the Beijing Natural Science Foundation under Grant L172049. The review of this article was coordinated by Dr. A. Al-Dweik. *(Corresponding author: Xiaoming Dai.)*

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Digital Object Identifier 10.1109/TVT.2020.3008151

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decentralized baseband processing scheme, each antenna cluster is associated with local computing fabrics that carry out the necessary RF and baseband processing tasks in a decentralized and parallel manner. Decentralized conjugate gradient (CG) [5] based and decentralized alternating direction method of multipliers (ADMM) [4], [6] based signal detection schemes were proposed to reduce the computational complexity of the linear decentralized MMSE detection. These decentralized detectors carry out the centralized processing by combining the matched filter output, Gram matrix, or some intermediate variables. Reference [7] proposed a decentralized coordinate descent (CD) based data detection to approximate the traditional decentralized MMSE detection for computational complexity reduction. The authors in [8]–[10] proposed a parallel detection relying on large-MIMO approximate message passing (LAMA) method. In these two methods, the centralized processing unit fuses the local estimations to form a global estimation via a weighted sum operation [7]–[10]. However, all these algorithms exhibit only comparable performance as the decentralized MMSE detection, and the bit error rate (BER) performance is unsatisfactory for the massive MIMO systems with large number of antenna clusters.

In this paper, we propose a novel decentralized signal detection scheme under the framework of expectation propagation algorithm (EPA) [12], [13] for the uplink massive MIMO systems. The EPA is a type of approximate message passing algorithm which approximates the symbol belief by a Gaussian probability density function (PDF) and simplifies the complicated message passing rules by mean and variance propagation. In the proposed decentralized scheme, the antennas at the BS are partitioned into multiple independent antenna clusters. In each cluster, the decentralized processing unit carries out local EPA detection according to the local CSI (received from the local CHEST module) and the local received signal. The centralized processing unit collects the means and variances propagated from each antenna cluster and computes the fused means and variances via the product principle of univariate Gaussian PDFs. The log-likelihood ratio (LLR) is updated iteratively based on the fused means and variances for decoding. To evaluate the feasibility of the proposed decentralized EPA architecture, we perform simulations for the uplink massive MIMO systems. Simulation results illustrate that the proposed decentralized EPA signal detection approaches the performance of the centralized one with marginal computational complexity overhead, and outperforms the state-of-the-art decentralized counterparts especially for large number of antenna clusters.

The remainder of this paper is organized as follows. Section II provides the decentralized uplink massive MIMO system model. Section III details the proposed decentralized EPA signal detection as well as the computational complexity analysis. Simulation results are presented in Section IV. Finally, conclusions are drawn in Section V.

Notation: Boldface uppercase letter $\mathbf{A}$ and lowercase letter $\mathbf{a}$ stand for matrices and column vectors, respectively. $\mathbf{A}^T$ represents the matrix transpose. $|\mathcal{A}|$ denotes the cardinality of set $\mathcal{A}$. E[.] denotes the expectation operation. $\mathbf{I}$ is the identity matrix. Finally, the complex Gaussian PDF with mean μ and variance σ^2 is represented by $\mathcal{CN}(x;\mu,\sigma^2) = (\pi\sigma^2)^{-1}\exp(-|x-\mu|^2/\sigma^2)$.

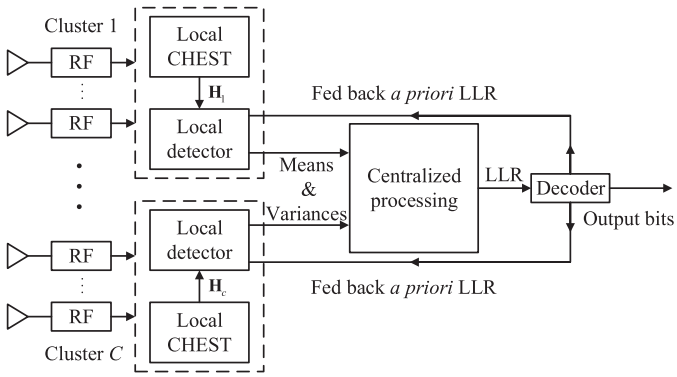


Fig. 1. Decentralized baseband processing based EPA receiver model.

II. System Model

We consider an uplink massive MIMO system where K single-antenna user equipments transmit signal to the BS with N ($N \gg K$) antenna elements in the same frequency resource simultaneously. The source bits $\mathbf{b}$ for all users are encoded by a channel encoder and permuted by a random interleaver, producing the coded and interleaved bits $\mathbf{c}$. The bit stream is mapped into a complex constellation $\mathcal{A}$ with cardinality $|\mathcal{A}| = 2^Q$ which results in the transmitted signal vector $\mathbf{x} = [x_1, x_2, \ldots, x_K]^T$. Each transmitted symbol x_k is associated with Q bits c_k^q where $q = 1, 2, \ldots, Q$. The average power of transmitted symbol is normalized to unit $\mathrm{E}[|x_k|^2] = 1$. The baseband input-output relation of the uplink channel is modeled as

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \tag{1}$$

where $\mathbf{y}$ is the received signal, $\mathbf{H}$ is the Rayleigh flat fading channel matrix whose entries are independent and identically distributed with zero mean and unit variance, and $\mathbf{n}$ denotes the complex additive white Gaussian noise (AWGN) vector with zero mean and covariance matrix $\sigma^2\mathbf{I}$. Perfect CSI is assumed to be available at the receiver and the signal-to-noise ratio (SNR) is defined as $\mathrm{SNR} = K/\sigma^2$.

In decentralized massive MIMO systems, N antennas at the BS are partitioned into C independent antenna clusters and the partition is assumed to be uniform. Each antenna cluster equips $M = N/C$ antennas and a decentralized processing unit which contains a local RF circuitry, local CHEST module, local detector, and relevant computing hardwares. By partitioning the channel matrix row-wise into blocks as $\mathbf{H} = [\mathbf{H}_1^T, \mathbf{H}_2^T, \ldots, \mathbf{H}_C^T]^T$, the received signal vector as $\mathbf{y} = [\mathbf{y}_1^T, \mathbf{y}_2^T, \ldots, \mathbf{y}_C^T]^T$, and the noise vector as $\mathbf{n} = [\mathbf{n}_1^T, \mathbf{n}_2^T, \ldots, \mathbf{n}_C^T]^T$, the received signal at the cth antenna cluster is denoted as

$$\mathbf{y}_c = \mathbf{H}_c\mathbf{x} + \mathbf{n}_c. \tag{2}$$

When the number of antenna clusters is $C = 1$, the decentralized massive MIMO system degenerates into the conventional centralized massive MIMO system.

Soft-input soft-output iterative detection and decoding (IDD) scheme adopts an iterative procedure in which the detector and decoder exchange the extrinsic information iteratively to extract the extrinsic LLR. The extrinsic LLR $L_e(c_k^q)$ of bit c_k^q generated

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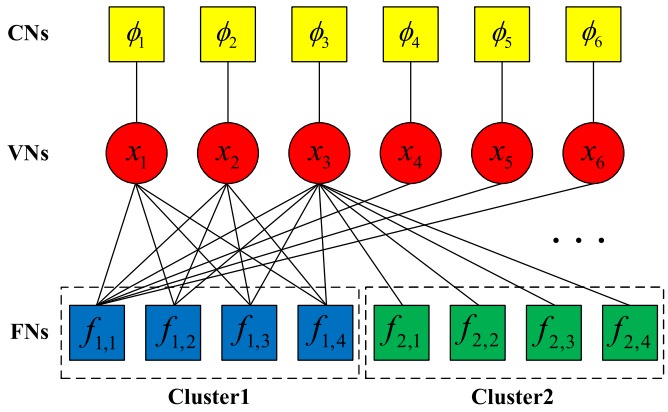


Fig. 2. Factor graph of decentralized EPA.

by the detector is computed as

$$L_e\left(c_k^q\right)=\ln\frac{p\left(c_k^q=1|\mathbf{y}\right)}{p\left(c_k^q=0|\mathbf{y}\right)}-L_a\left(c_k^q\right), \tag{3}$$

where $L_a(c_k^q)$ is the *a priori* LLR fed back from the decoder. The block diagram of the decentralized baseband processing based EPA receiver model is illustrated in Fig. 1.

III. Decentralized Signal Detection Based on Expectation Propagation Algorithm

A. Factor Graph Model

The signal processing by EPA can be viewed as an instance of the sum-product algorithm [12] which operates the message passing in a factor graph. The joint distribution $p(\mathbf{x},\mathbf{y})$ can be factorized into [14], [15]

$$p\left(\mathbf{x},\mathbf{y}\right)=p\left(\mathbf{x}\right)p\left(\mathbf{y}|\,\mathbf{x}\right)=\prod_k p\left(x_k\right)\prod_n f_n\left(y_n|\,\mathbf{x}\right), \tag{4}$$

where $p(\mathbf{x})$ denotes the *a priori* distribution of $\mathbf{x}$, and the likelihood function $f_n(y_n|\mathbf{x})$ is computed by

$$f_n\left(y_n|\,\mathbf{x}\right)=\frac{1}{\pi\sigma^2}\exp\left(-\frac{\left|y_n-\sum_k h_{n,k}x_k\right|^2}{\sigma^2}\right), \tag{5}$$

where the nth row and kth column element in the CSI matrix $\mathbf{H}$ is denoted by $h_{n,k}$. The factorization in (4) can be represented by the factor graph. A factor graph example of decentralized EPA is illustrated in Fig. 2, where the antenna configuration is $N\times K=8\times 6$ and $C=2$. The variable nodes (VNs), factor nodes (FNs), and cloning nodes (CNs) represent the users, likelihood functions, and constellation mapping constraints, respectively. The inner iteration denotes the message passing between VNs and FNs iteratively, and the outer iteration represents the IDD procedure. The number of inner and outer iterations are denoted by i and t, respectively. The maximum number of inner and outer iterations are denoted by $I_{\max}$ and $T_{\max}$, respectively.

B. Decentralized Expectation Propagation Algorithm

The key idea of the EPA is to approximate a (often intractable) distribution p with another tractable distribution q which is constrained to lie in a family of simple distribution set Φ through distribution projection. Let Φ be some set of distributions and the projection of a distribution p into the set Φ is represented as [15]

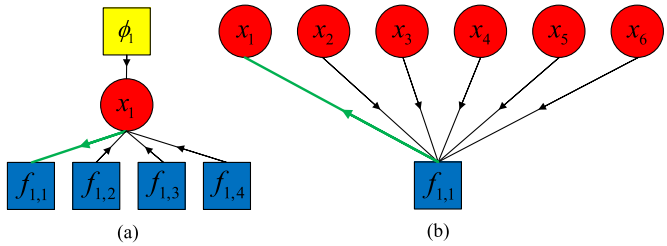


Fig. 3. Message passing rules between VN and FN. (a) Message passing from VN to FN. (b) Message passing from FN to VN.

$$\mathrm{Proj}_{\Phi}(p)=\arg\min_{q\in\Phi}D_{\mathrm{KL}}\left(p|\,|q\right), \tag{6}$$

where $D_{\mathrm{KL}}(p||q)$ denotes the Kullback-Leibler (KL) divergence which is an asymmetric measure of dissimilarity between two probability distributions p and q.

In the decentralized processing unit, we consider the decentralized EPA signal detection in the cth antenna cluster with M antennas and K users. The mth element in the local received signal $\mathbf{y}_c$ is $y_{c,m}$ while the mth row and kth column element in the local CSI matrix $\mathbf{H}_c$ is $h_{c,mk}$. Let $\mu^{i,t}_{x_k\to f_{c,m}}(x_k)$ denote the message passing from VN x_k to FN $f_{c,m}$ at the ith inner iteration and the tth outer iteration of the cth antenna cluster, and $\mu^{i,t}_{f_{c,m}\to x_k}(x_k)$ represent the message passing in the opposite direction. The detailed description of message passing rules between VN and FN is depicted in Fig. 3. The message passing rules are expressed as [12], [14]

$$\mu^{i,t}_{x_k\to f_{c,m}}\left(x_k\right)=\mu^{t}_{\phi_k\to x_k}\left(x_k\right)\prod_{m'\neq m}\mu^{i-1,t}_{f_{c,m'}\to x_k}\left(x_k\right), \tag{7}$$

$$\mu^{i,t}_{f_{c,m}\to x_k}\left(x_k\right)=\sum_{\mathbf{x}\backslash x_k}f_{c,m}\left(y_{c,m}|\,\mathbf{x}\right)\prod_{k'\neq k}\mu^{i,t}_{x_{k'}\to f_{c,m}}\left(x_{k'}\right), \tag{8}$$

where $\mathbf{x}\backslash x_k$ represents all the enumerations of $\mathbf{x}$ except for x_k, and $\mu^{t}_{\phi_k\to x_k}(x_k)=p(x_k)$ is computed as

$$\mu^{t}_{\phi_k\to x_k}\left(x_k\right)=\prod_q\frac{\exp\left(c_k^qL_a^t\left(c_k^q\right)\right)}{1+\exp\left(L_a^t\left(c_k^q\right)\right)}, \tag{9}$$

which denotes the *a priori* probability of x_k fed back from the decoder at the tth outer iteration. The *a priori* message $\mu^{t}_{\phi_k\to f_k}(x_k)$ is shared by all the antenna clusters, and $L_a^0(c_k^q)=0$ is initialized since there is no *a priori* information fed back from the decoder at the initial iteration. The symbol belief in the cth antenna cluster, i.e., the *a posteriori* probability of x_k, is calculated as [14]

$$p^{i,t}\left(x_k\right)=\frac{1}{Z}\mu^{t}_{\phi_k\to x_k}\left(x_k\right)\prod_m\mu^{i-1,t}_{f_{c,m}\to x_k}\left(x_k\right), \tag{10}$$

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where Z is the normalization constant. According to (7) and (10), the message $\mu^{i,t}_{x_k\to f_{c,m}}(x_k)$ is rewritten as

$$\mu^{i,t}_{x_k\to f_{c,m}}(x_k)=\frac{Zp^{i,t}(x_k)}{\mu^{i-1,t}_{f_{c,m}\to x_k}(x_k)}. \tag{11}$$

Since the symbols are taken on values in the discrete constellation set $\mathcal{A}$, the computation of $\mu^{i,t}_{f_{c,m}\to x_k}(x_k)$ in (8) requires exponential time to marginalize out the vector $\mathbf{x}\backslash x_k$. To reduce the computational complexity of message passing rules, the messages $\mu^{i,t}_{x_k\to f_{c,m}}(x_k)$ and $\mu^{i,t}_{f_{c,m}\to x_k}(x_k)$ are approximated as complex Gaussian PDFs as $\mathcal{CN}(x_k;x^{i,t}_{x_k\to f_{c,m}},v^{i,t}_{x_k\to f_{c,m}})$ and $\mathcal{CN}(x_k;x^{i,t}_{f_{c,m}\to x_k},v^{i,t}_{f_{c,m}\to x_k})$, respectively. The symbol belief $p^{i,t}(x_k)$ is also approximated as a continuous complex Gaussian PDF, which results in an approximation $\hat{p}^{i,t}(x_k)=\mathcal{CN}(x_k;\hat{x}^{i,t}_k,\hat{v}^{i,t}_k)$. The mean and variance of the approximate symbol belief $\hat{p}^{i,t}(x_k)$ are obtained by moment matching [14]

$$\hat{x}^{i,t}_k=\sum_{\alpha\in\mathcal{A}}\alpha p^{i,t}(x_k=\alpha), \tag{12}$$

$$\hat{v}^{i,t}_k=\sum_{\alpha\in\mathcal{A}}\left|\alpha-\hat{x}^{i,t}_k\right|^2 p^{i,t}(x_k=\alpha), \tag{13}$$

where α is the constellation symbol in the set $\mathcal{A}$. Since the Gaussian distribution can be fully characterized by the mean and variance, the complicated message passing rule in (11) can be simplified into mean and variance parameter update after the Gaussian approximations are adopted. Based on (11) and the approximate symbol belief we obtain

$$\mu^{i,t}_{x_k\to f_{c,m}}(x_k)\propto\frac{p^{i,t}(x_k)}{\mu^{i-1,t}_{f_{c,m}\to x_k}(x_k)}\approx\frac{\hat{p}^{i,t}(x_k)}{\mu^{i-1,t}_{f_{c,m}\to x_k}(x_k)}, \tag{14}$$

and it is denoted as

$$\mathcal{CN}\left(x_k;x^{i,t}_{x_k\to f_{c,m}},v^{i,t}_{x_k\to f_{c,m}}\right)\propto\frac{\mathcal{CN}\left(x_k;\hat{x}^{i,t}_k,\hat{v}^{i,t}_k\right)}{\mathcal{CN}\left(x_k;x^{i-1,t}_{f_{c,m}\to x_k},v^{i-1,t}_{f_{c,m}\to x_k}\right)}. \tag{15}$$

The means and variances propagated from VNs to FNs are thus computed respectively as [16]

$$v^{i,t}_{x_k\to f_{c,m}}=\left(\frac{1}{\hat{v}^{i,t}_k}-\frac{1}{v^{i-1,t}_{f_{c,m}\to x_k}}\right)^{-1}, \tag{16}$$

$$x^{i,t}_{x_k\to f_{c,m}}=v^{i,t}_{x_k\to f_{c,m}}\left(\frac{\hat{x}^{i,t}_k}{\hat{v}^{i,t}_k}-\frac{x^{i-1,t}_{f_{c,m}\to x_k}}{v^{i-1,t}_{f_{c,m}\to x_k}}\right), \tag{17}$$

where the initial value $x^{0,t}_{f_{c,m}\to x_k}=0$ and $v^{0,t}_{f_{c,m}\to x_k}=+\infty$ can be set at the initial inner iteration. Since $\mu^{i,t}_{f_{c,m}\to x_k}(x_k)$ has been approximated as a complex Gaussian PDF, the calculation of (8) can be solved by integration [14], [17]

$$\begin{aligned}\mu^{i,t}_{f_{c,m}\to x_k}(x_k)&=\int_{\mathbf{x}\backslash x_k}f_{c,m}(y_{c,m}|\mathbf{x})\prod_{k'\neq k}\mu^{i,t}_{x_{k'}\to f_{c,m}}(x_{k'})\\&=\mathcal{CN}\left(x_k;x^{i,t}_{f_{c,m}\to x_k},v^{i,t}_{f_{c,m}\to x_k}\right),\end{aligned} \tag{18}$$

where the means and variances propagated from FNs to VNs are computed respectively as

$$x^{i,t}_{f_{c,m}\to x_k}=\frac{1}{h_{c,mk}}\left(y_{c,m}-\sum_{k'\neq k}h_{c,mk'}x^{i,t}_{x_{k'}\to f_{c,m}}\right), \tag{19}$$

$$v^{i,t}_{f_{c,m}\to x_k}=\frac{1}{|h_{c,mk}|^2}\left(\sum_{k'\neq k}|h_{c,mk'}|^2v^{i,t}_{x_{k'}\to f_{c,m}}+\sigma^2\right). \tag{20}$$

The means and variances are updated iteratively between VNs and FNs until the maximum number of inner iterations is reached. After inner iterations, the VNs collect the message passed from the FNs in each antenna cluster, and the message passed from VNs to CNs is computed as

$$\mu^{t,c}_{x_k\to\phi_k}(x_k)=\prod_m\mu^{I_{\max},t}_{f_{c,m}\to x_k}(x_k), \tag{21}$$

which is the multiplication of all the incoming messages $\mu^{I_{\max},t}_{f_{c,m}\to x_k}(x_k)$. The message $\mu^{t,c}_{x_k\to\phi_k}(x_k)$ is modeled as a complex Gaussian PDF $\mathcal{CN}(x_k;\tilde{x}^{t,c}_k,\tilde{v}^{t,c}_k)$ whose mean and variance are calculated respectively as [16]

$$\tilde{v}^{t,c}_k=\left(\sum_m\frac{1}{v^{I_{\max},t}_{f_{c,m}\to x_k}}\right)^{-1}, \tag{22}$$

$$\tilde{x}^{t,c}_k=\tilde{v}^{t,c}_k\left(\sum_m\frac{x^{I_{\max},t}_{f_{c,m}\to x_k}}{v^{I_{\max},t}_{f_{c,m}\to x_k}}\right). \tag{23}$$

Note that the means and variances of the multiplicative message $\prod_m\mu^{i-1,t}_{f_{c,m}\to x_k}(x_k)$ in (10) can be calculated based on (22) and (23) by replacing the superscript $I_{\max}$ with $i-1$.

In the centralized processing unit, the messages from all the C antenna clusters are fused by the proposed message fusion rule. Integrating all the means $\tilde{x}^{t,c}_k$ and variances $\tilde{v}^{t,c}_k$ propagated from each antenna cluster, the fused message $\mu^t_{x_k\to\phi_k}(x_k)=\mathcal{CN}(x_k;\tilde{x}^t_k,\tilde{v}^t_k)$ are obtained based on the multiplication of all the incoming message $\mu^{t,c}_{x_k\to\phi_k}(x_k)$, and the mean and variance are computed respectively as [16]

$$\tilde{v}^t_k=\left(\sum_c\frac{1}{\tilde{v}^{t,c}_k}\right)^{-1}=\left(\sum_c\sum_m\frac{1}{v^{I_{\max},t}_{f_{c,m}\to x_k}}\right)^{-1}, \tag{24}$$

$$\tilde{x}^t_k=\tilde{v}^t_k\left(\sum_c\frac{\tilde{x}^{t,c}_k}{\tilde{v}^{t,c}_k}\right)=\tilde{v}^t_k\left(\sum_c\sum_m\frac{x^{I_{\max},t}_{f_{c,m}\to x_k}}{v^{I_{\max},t}_{f_{c,m}\to x_k}}\right). \tag{25}$$

Based on the fused message $\mu^t_{x_k\to\phi_k}(x_k)$, the updated symbol belief $\beta^t(x_k)$ is generated as [14]

$$\beta^t(x_k)=\frac{1}{Z}\mu^t_{\phi_k\to x_k}(x_k)\,\mathcal{CN}\left(x_k;\tilde{x}^t_k,\tilde{v}^t_k\right), \tag{26}$$

where Z is the normalization constant. The extrinsic LLR generated by the decentralized EPA detector is computed as

$$L^t_e(c^q_k)=\ln\frac{\sum_{x_k\in\mathcal{A}^1_q}\beta^t(x_k)}{\sum_{x_k\in\mathcal{A}^0_q}\beta^t(x_k)}-L^t_a(c^q_k), \tag{27}$$

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TABLE I
COMPUTATIONAL COMPLEXITY

Decentralized processing (each inner iteration)	Message from VNs to FNs	$(8M+3\times 2^Q+3Q+5)K$
	Message from FNs to VNs	$5MK$
Centralized processing	Computation of the extrinsic LLR	$(2N+7)K$
Total complexity of Decentralized EPA	$T_{\max}CI_{\max}(13M+3\times 2^Q+3Q+5)K$ $+T_{\max}(2N+7)K$	
Decentralized LAMA	$(N+CI_{\max})K^2+(CI_{\max}(2^{Q+2}+4)+N+C)K$ $+3CI_{\max}\times 2^Q$	
Decentralized CD	$(2I_{\max}(N+C)+N+2C)K$	
Decentralized MMSE	$\frac{7}{3}CK^3+2NK^2+(N+\frac{2}{3}C+1)K+2$	
Centralized EPA	$T_{\max}I_{\max}(13M+3\times 2^Q+3Q+5)K$ $+T_{\max}(2N+7)K$	

where $\mathcal{A}_q^0$ and $\mathcal{A}_q^1$ denote the subset of constellation points with the qth bit being 0 and 1, respectively. The detector and decoder deliver extrinsic information iteratively to refine the extrinsic LLR. At the final outer iteration, the decoder carries out hard decision to determine the decoded bits. The overall procedure of the proposed decentralized EPA signal detection for the uplink massive MIMO systems is summarized in Algorithm 1. Note that when the number of antenna clusters is $C=1$, the proposed decentralized EPA detection is equivalent to the classical EPA one for the centralized massive MIMO systems. To further improve the performance of the proposed scheme, the successive or parallel interference cancellation can be implemented in each antenna cluster parallelly based on the local CSI and reconstructed symbols.

Algorithm 1: Decentralized EPA Signal Detection.

Input: $\mathbf{y}_c$, $\mathbf{H}_c$, σ^2, $L_a^0(c_k^q)=0$, $I_{\max}$, $T_{\max}$
Initialization: $x_{f_{c,m}\to x_k}^{0,t}=0$, $v_{f_{c,m}\to x_k}^{0,t}=+\infty$
Decentralized processing in each cluster: $c=1,\ldots,C$
for $i=1,\cdots,I_{\max}$ **do** (inner iteration)
  for $k=1,\ldots,K$ **do** (message from VNs to FNs)
    $\mu_{\phi_k\to x_k}^t(x_k)=\prod_q \frac{\exp(c_k^q L_a^t(c_k^q))}{1+\exp(L_a^t(c_k^q))}$
    $p^{i,t}(x_k)=\frac{1}{Z}\mu_{\phi_k\to x_k}^t(x_k)\prod_m \mu_{f_{c,m}\to x_k}^{i-1,t}(x_k)$
    $\hat{x}_k^{i,t}=\sum_{\alpha\in\mathcal{A}}\alpha p^{i,t}(x_k=\alpha)$
    $\hat{v}_k^{i,t}=\sum_{\alpha\in\mathcal{A}}|\alpha-\hat{x}_k^{i,t}|^2 p^{i,t}(x_k=\alpha)$
    for $m=1,\ldots,M$ **do**
      $v_{x_k\to f_{c,m}}^{i,t}=\left(\frac{1}{\hat{v}_k^{i,t}}-\frac{1}{v_{f_{c,m}\to x_k}^{i-1,t}}\right)^{-1}$
      $x_{x_k\to f_{c,m}}^{i,t}=v_{x_k\to f_{c,m}}^{i,t}\left(\frac{\hat{x}_k^{i,t}}{\hat{v}_k^{i,t}}-\frac{x_{f_{c,m}\to x_k}^{i-1,t}}{v_{f_{c,m}\to x_k}^{i-1,t}}\right)$
    end for
  end for
  for $m=1,\ldots,M$ **do** (message from FNs to VNs)
    for $k=1,\ldots,K$ **do**
      $x_{f_{c,m}\to x_k}^{i,t}=\frac{y_{c,m}-\sum_{k'\neq k}h_{c,mk'}x_{x_{k'}\to f_{c,m}}^{i,t}}{h_{c,mk}}$
      $v_{f_{c,m}\to x_k}^{i,t}=\frac{\sum_{k'\neq k}|h_{c,mk'}|^2 v_{x_{k'}\to f_{c,m}}^{i,t}+\sigma^2}{|h_{c,mk}|^2}$
    end for
  end for
end for
Centralized processing: message fusion
for $k=1,\ldots,K$ **do** (compute extrinsic LLR)
  $\tilde{v}_k^t=\left(\sum_c\sum_m \frac{1}{v_{f_{c,m}\to x_k}^{I_{\max},t}}\right)^{-1}$
  $\tilde{x}_k^t=\tilde{v}_k^t\left(\sum_c\sum_m \frac{x_{f_{c,m}\to x_k}^{I_{\max},t}}{v_{f_{c,m}\to x_k}^{I_{\max},t}}\right)$
  $\beta^t(x_k)=\frac{1}{Z}\mu_{\phi_k\to x_k}^t(x_k)\mathcal{CN}(x_k;\tilde{x}_k^t,\tilde{v}_k^t)$
  for $q=1,\ldots,Q$ **do**
    $L_e^t(c_k^q)=\ln\frac{\sum_{x_k\in\mathcal{A}_q^1}\beta^t(x_k)}{\sum_{x_k\in\mathcal{A}_q^0}\beta^t(x_k)}-L_a^t(c_k^q)$
  end for
end for
Executing $T_{\max}$ **outer iterations.**
Output: $L_e^{T_{\max}}(c_k^q)$

C. Computational Complexity Analysis

The computational complexity analysis of the proposed decentralized EPA signal detection in terms of the required complex multiplication number is provided in this subsection. The computational complexity of the decentralized EPA detection at each inner iteration can be divided into three parts. (a) Message from VNs to FNs: $(8M+3\times 2^Q+3Q+5)K$. (b) Message from FNs to VNs: $5MK$. (c) Computation of the extrinsic LLR according to the fused message: $(2N+7)K$. The overall computational complexity is computed as $T_{\max}I_{\max}C(13M+3\times 2^Q+3Q+5)K+T_{\max}(2N+7)K$. The computational complexity of the decentralized LAMA [8], [9], CD [7], MMSE, and centralized EPA methods are provided. The number of iterations for the decentralized LAMA and CD detections are denoted by $I_{\max}$. The overall computational complexity is listed in Table I.

IV. SIMULATION RESULTS

In this section, simulation results of the proposed decentralized EPA signal detection and the state-of-the-art decentralized detection schemes [7]–[10] are evaluated. We consider two uplink massive MIMO system configurations with $N\times K=64\times 16$ and $N\times K=128\times 16$. The antennas at the BS are partitioned into $C=2$ or $C=4$ antenna clusters uniformly. The convolutional code with $1/2$ code rate and $[133_o 171_o]$ polynomial is utilized. Quadrature phase shift keying (QPSK) and 16 quadrature amplitude modulation (QAM) modulation schemes are adopted. The number of inner and outer iterations for EPA, the number of iterations for CD and LAMA are all set to be 3.

A. BER Performance

Fig. 4 and Fig. 5 illustrate the BER performance of the decentralized signal detection algorithms with $N\times K=64\times 16$, and the number of clusters are $C=2$ and $C=4$, respectively. For QPSK modulation, the decentralized CD detection achieves almost identical performance as the decentralized MMSE detection while the decentralized LAMA algorithm suffers slight

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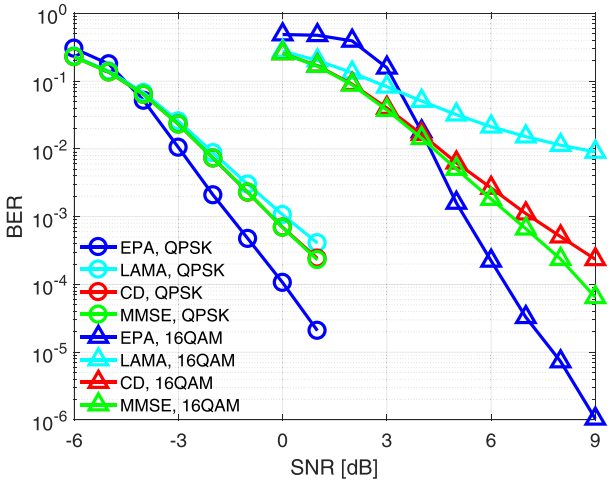


Fig. 4. BER versus SNR for $N \times K = 64 \times 16$ and $C = 2$.

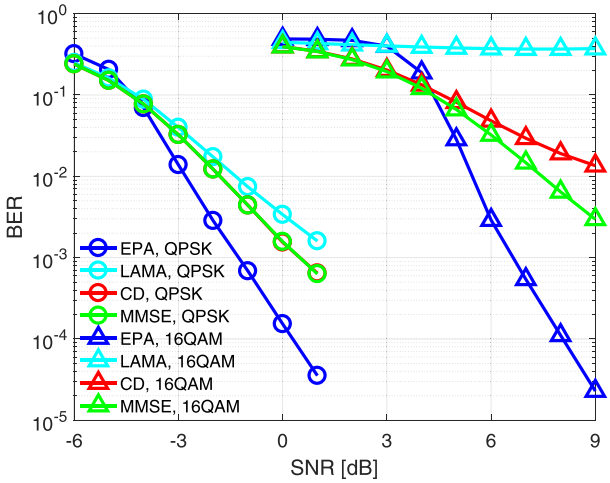


Fig. 5. BER versus SNR for $N \times K = 64 \times 16$ and $C = 4$.

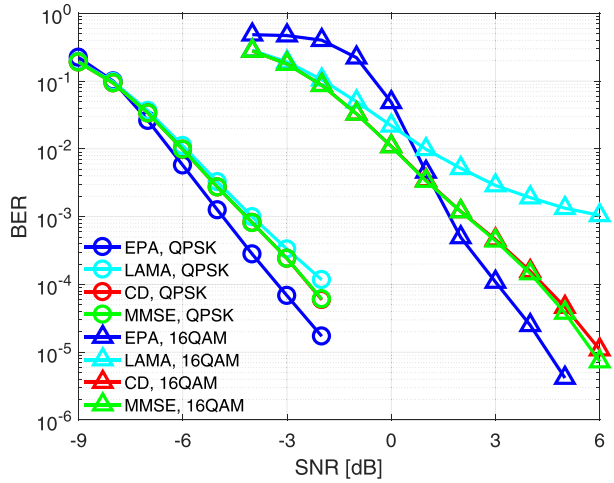


Fig. 6. BER versus SNR for $N \times K = 128 \times 16$ and $C = 2$.

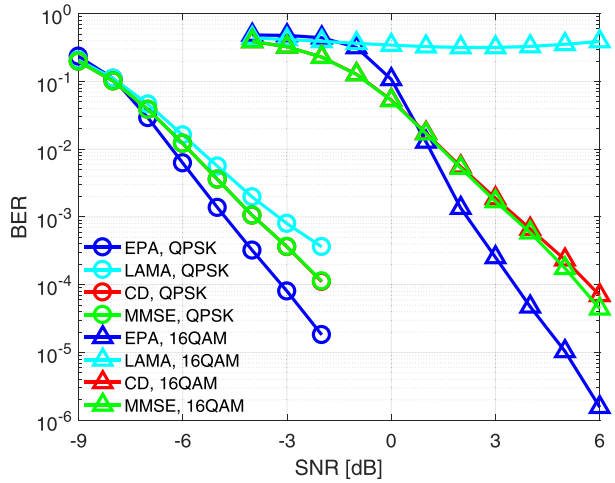


Fig. 7. BER versus SNR for $N \times K = 128 \times 16$ and $C = 4$.

performance degradation. The proposed decentralized EPA detection scheme provides about 1.2 dB and 1.8 dB performance improvements over the decentralized MMSE detection at BER $= 1 \times 10^{-3}$ for $C = 2$ and $C = 4$, respectively. For 16-QAM modulation, the proposed decentralized EPA detection achieves greater performance gain at the high-SNR region compared with the QPSK modulation. The decentralized CD method is unable to approach decentralized MMSE detection, and the decentralized LAMA even fails to converge when the number of clusters is $C = 4$. Similar scenarios are observed in $N \times K = 128 \times 16$ configuration as shown in Fig. 6 and Fig. 7. The performance advantage of the proposed decentralized EPA method is reduced. For QPSK modulation, the performance gaps between the proposed decentralized EPA detection and decentralized MMSE detection are about 0.7 dB and 0.9 dB at BER $= 1 \times 10^{-3}$ for $C = 2$ and $C = 4$, respectively. The decentralized CD detection achieves similar performance as the MMSE counterpart with 16-QAM modulation.

Fig. 8 and Fig. 9 illustrate the BER performance for $N \times K = 64 \times 16$ configuration with QPSK and 16-QAM modulation schemes, respectively. The decentralized signal detections suffer from larger performance losses compared with the conventional centralized counterparts with the increase of cluster number. This is attributed to the fact that the smaller number of antenna resources in each cluster fails to satisfy the condition $M \gg K$. For QPSK modulation, the proposed decentralized EPA detection suffers less performance loss compared with the centralized counterpart with the increase of cluster number. However, the performance degradation caused by the decentralized processing is apparent for the other decentralized detections. For 16-QAM modulation, the performance losses for the decentralized detectors are more prominent, however, the proposed decentralized EPA detection with $C = 2$ still obtains near-centralized performance. Fig. 8 and Fig. 9 illustrate that the centralized LAMA detection exhibits similar performance as the centralized MMSE detection, but the noticeable performance degradation is observed for the decentralized LAMA detection with high order modulation. It is attributed to the non-optimal message fusion rule and slow convergence rate of the decentralized LAMA detection. Soft-input soft-output IDD scheme can be adopted to improve the performance of the decentralized LAMA detection, however, it is necessary to design a suitable

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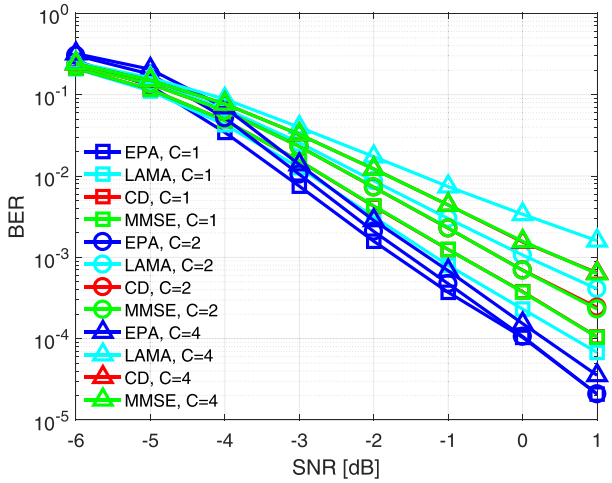


Fig. 8. BER versus SNR for $N \times K = 64 \times 16$ and QPSK.

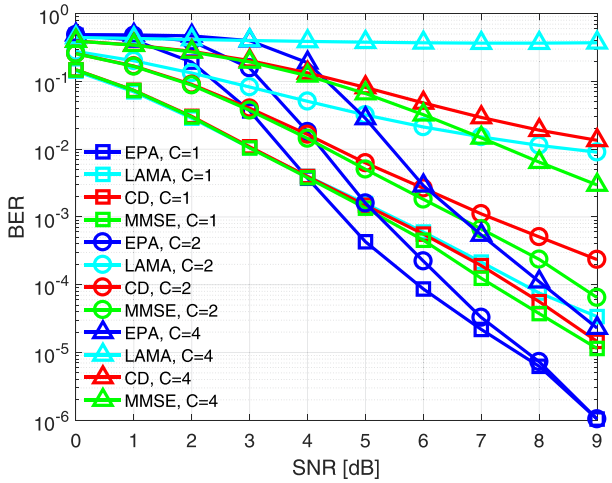


Fig. 9. BER versus SNR for $N \times K = 64 \times 16$ and 16-QAM.

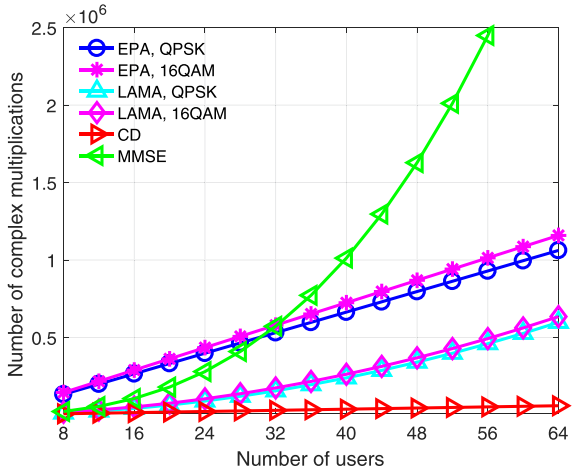


Fig. 10. Computational complexity comparison of the decentralized signal detections for $N = 128$ and $C = 4$.

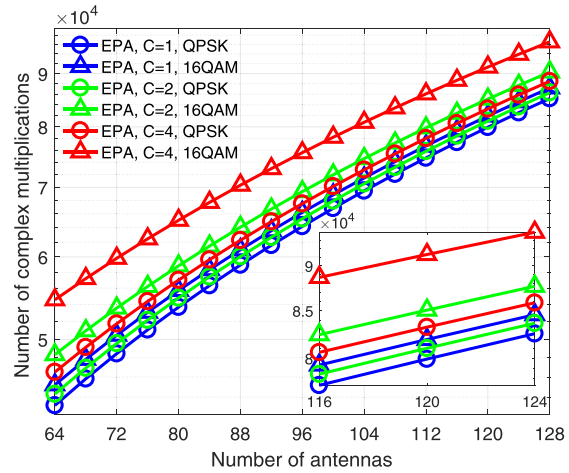


Fig. 11. Computational complexity comparison of the centralized and decentralized EPA signal detections for $K = 16$, $T_{\max} = 1$, and $I_{\max} = 3$.

message fusion rule for soft-input soft-output processing. CD method is a classical iteration algorithm which can reduce the computational complexity of the complicated matrix inversion in MMSE detection, and the simulation results illustrate that the BER performance of the decentralized CD detection approaches the MMSE counterpart. The proposed decentralized EPA signal detection scheme exhibits the best performance which is benefited from the fast convergence rate of the EPA and the elaborately designed message fusion rule.

B. Computational Complexity

Numerical results of the computational complexity are illustrated in Fig. 10 and Fig. 11. The computational complexity comparison between the decentralized EPA detection and the other decentralized methods is depicted in Fig. 10, where the number of antennas and clusters are $N = 128$ and $C = 4$, respectively. The decentralized MMSE shows the highest computational complexity while the complexity of CD is the lowest, but the BER performance of the CD method only approaches the MMSE detection. The proposed decentralized EPA detection exhibits lower computational complexity than the decentralized MMSE detection with large number of users, and it achieves a better tradeoff between system performance and computational complexity. Fig. 11 illustrates the computational complexity comparison of the centralized and decentralized EPA detections, where the number of users, outer iterations, and inner iterations are $K = 16$, $T_{\max} = 1$, and $I_{\max} = 3$, respectively. As shown in this figure, the proposed decentralized EPA detection exhibits marginal computational complexity overhead compared with the centralized counterpart. For example, when $C = 2$ with 16-QAM modulation and $C = 4$ with QPSK modulation for $N = 128$ antennas configuration, the computational complexity increments compared with the centralized EPA are 3.6% and 3.9%, respectively.

V. Conclusion

In this paper, we proposed a novel decentralized EPA signal detection for the uplink massive MIMO systems. This scheme

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contains of a decentralized processing unit which operates the local CHEST and local EPA signal detection, as well as a centralized processing unit which carries out message fusion and LLR refining for decoding. Simulation results illustrated that the proposed decentralized EPA signal detection achieves similar performance as that of the centralized counterpart with marginal computational complexity overhead, and outperforms the state-of-the-art decentralized signal detection schemes.

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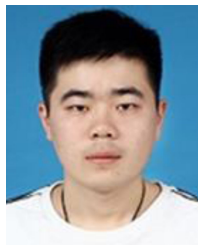
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